

TITLE: DATA AGENT PROJECT

Author: Violet Lesunyane

Student Numbver: 2805001682

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1. INTRODUCTION

This project was developed as a test project to evaluate my ability to create a functional Data Agent using Databricks. The purpose was to assess my understanding of the project requirements, my ability to follow instructions, and my capability to successfully complete the assigned task.

2. OBJECTIVE

The objective of this project is to create a functional Retail Sales Data Agent that uses Artificial Intelligence to help business owners to better understand their companies through data-driven insights.

The Data Agent must b designed to:

- Analyse customer behaviour
- Identify why certain products perform better than others
- Detect underperforming products
- Improve sales performance analysis
- Support smarter business decision-making

The ultimate goal of the project is to help businesses use data and AI to make more informed strategic decisions, improve customer understanding, and drive business growth.

3. DATA DESCRIPTION

The provided data had the following data for me to Extract, Transform and Load on Genie Space:

Transaction ID

Transaction Date

Customer ID

Gender

Age

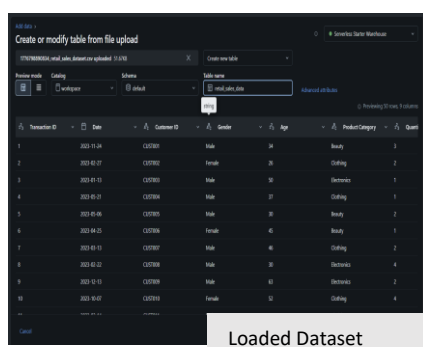
Product Category

Quantity

Price per Unit

Total Amount

4. EXTRACT > TRANSFORM > LOADING



Loaded Dataset

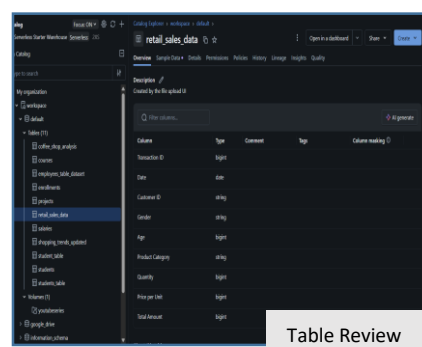


Table Review

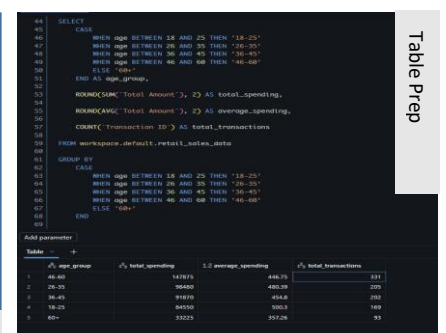
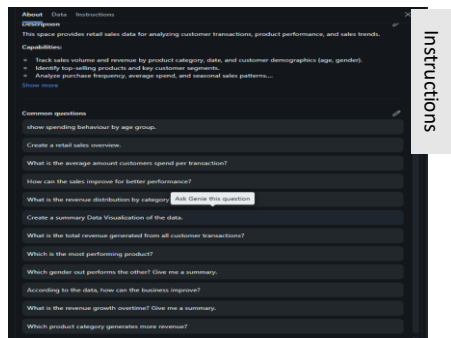
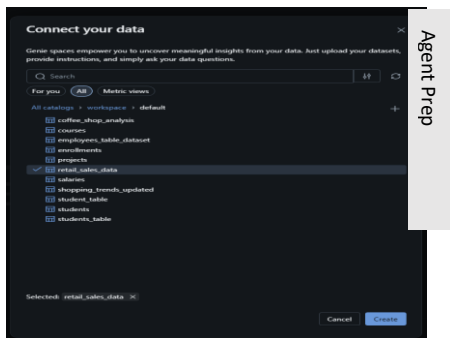


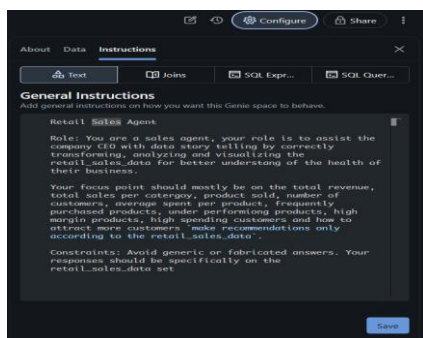
Table Prep

```
44 SELECT
45 CASE
46   WHEN age BETWEEN 18 AND 25 THEN '18-25'
47   WHEN age BETWEEN 26 AND 35 THEN '26-35'
48   WHEN age BETWEEN 36 AND 45 THEN '36-45'
49   WHEN age BETWEEN 46 AND 60 THEN '46-60'
50   ELSE 'Other'
51 END AS age_group,
52 ROUND(SUM('Total Amount'), 2) AS total_spending,
53 ROUND(AVG('Total Amount'), 2) AS average_spending,
54 COUNT('transaction id') AS total_transactions
55 FROM workspace.default.retail_sales_data
56 GROUP BY
57   age_group
58 END
```

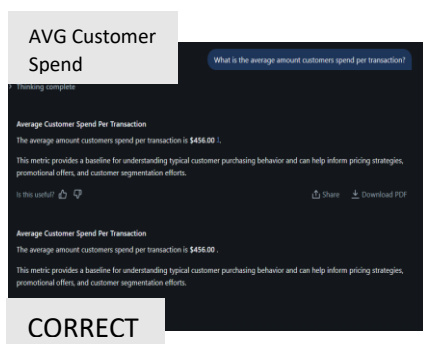
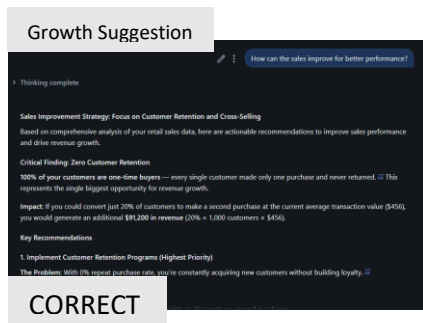
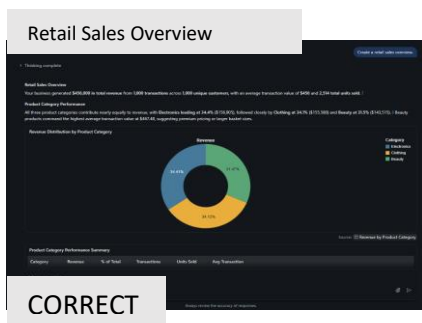
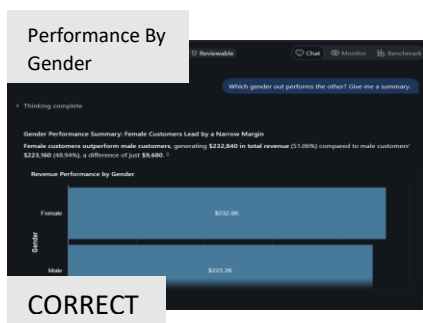
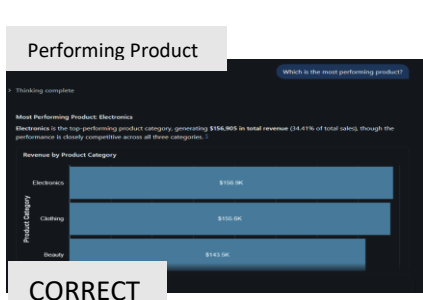
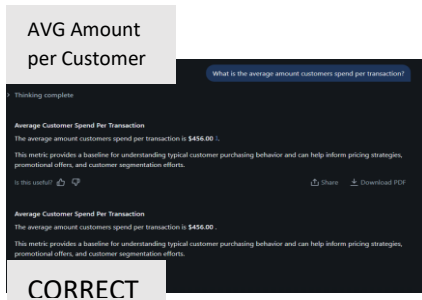
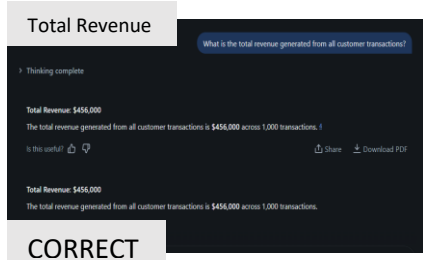
age_group	total_spending	average_spending	total_transactions
18-25	147971	846.78	331
26-35	104509	460.29	229
36-45	91920	414.82	220
46-60	84550	500.5	169
Other	33225	257.26	93



5. AGENT INSTRUCTIONS



6. 10 QUESTIONS



Revenue Distribution by Category



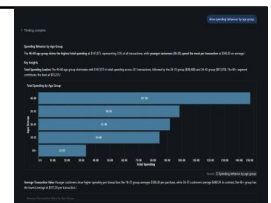
Across all the questions I uploaded to the AI Agent, nine returned correct outputs, with the exception of the customer spending behaviour query. To resolve this issue, I reviewed the original query, copied it, and ran it in the Genie Space SQL environment, which produced more accurate results.

7. VALIDATION OF 3 ANSWERS



```
SELECT
    C.Category,
    SUM(T.Total_Amount) AS Total_Amount,
    COUNT(DISTINCT T.Customer_ID) AS Customer_Count,
    ROUND(AVG(T.Total_Amount / COUNT(DISTINCT T.Customer_ID)), 2) AS Average_Transaction_Value
FROM Transactions T
GROUP BY C.Category
```

Category	Total_Amount	Customer_Count	Average_Transaction_Value
Electronics	147875	1000	147.875
Clothing	170100	1000	170.1
Beauty	148025	1000	148.025



Questions Cross Check

8. KEY INSIGHTS

The retail business is operationally functional but strategically underperforming with significant untapped growth potential.

8.1 The Core Problem

The business generated \$456,000 in revenue from 1,000 customers, but the critical weakness is zero customer retention every customer made exactly one purchase. This one-time buyer model severely limits growth and profitability, as acquiring new customers is typically 5-25x more expensive than retaining existing ones.

8.2 What's Working

Balanced Product Portfolio: All three categories (Electronics, Clothing, Beauty) contribute roughly equally to revenue (31-34% each), demonstrating diversified income streams and reducing dependency risk.

Strong Average Transaction Value: At \$456 per transaction, customers are willing to spend meaningfully when they do purchase, with Beauty products commanding the highest value at \$467.481.

Valuable Core Demographic: The 46-60 age group represents your most reliable customer base, contributing 32% of total revenue (\$147,875).

8.3 What's Not Working

No Customer Loyalty: The complete absence of repeat purchases indicates fundamental issues with customer experience, engagement, or retention strategy.

Revenue Volatility: Monthly sales swing wildly from \$23,620 to \$53,150, suggesting inconsistent marketing, inventory management, or seasonal planning.

Underutilized High-Value Segments: Young adults (18-25) spend the most per transaction (\$500.30) but represent only 17% of customers, while seniors (60+) are barely engaged with just 93 customers.

9. BUSINESS RECOMMENDATIONS

Immediate Priority: Implement a customer retention program (loyalty rewards, email marketing, personalized follow-ups) to convert one-time buyers into repeat customers. Even a modest 20% repeat purchase rate could add at least \$90,000+ in annual revenue without new customer acquisition costs.

Growth Opportunity: Target the 18-25 demographic with digital-first marketing and expand senior outreach to capture underserved high-potential segments.

Stabilization Need: Address monthly revenue volatility through consistent promotional calendars, better inventory planning, and understanding what drove the companies best and worst months.

In Conclusion

The business is viable, with solid fundamentals but is operating at a fraction of potential capacity due to the absence of customer retention. The business is currently in "acquisition-only mode" when it should be in "acquire and retain mode." Shifting focus from purely transactional relationships to building customer loyalty will be the difference between stagnant performance and sustainable growth. The data shows customers are willing to spend you just need to give them reasons to come back.

10. CONCLUSION

10.1 What worked

Querying the data first before connecting it onto the Genie Space assists with better question outcomes.

10.2 What was hard

I did not struggle much but I would still like to master my coding language some more for even better results.

10.3 What you would change

Doubting my self